

If pictures are needed/relevant, please provide them with your solutions.

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The x - y plane ($z = 0$) is covered uniformly with electric charge, with surface charge density σ (i.e., any area A of the plane carries charge σA). Assume σ to be positive.

(a) **[3 pts.]** By symmetry, what should be the direction of the electric field $\mathbf{E}$ at the point P ? What should be the direction of the electric field at any other point with positive z -coordinate? What about points with negative z -coordinate?

- (b) **[2+2 pts.]** Consider an infinitely long strip of the charged plane along the x -direction, running from $x = -\infty$ to $x = +\infty$, parallel to the x -axis. Let the strip be infinitesimally thin, located between $y = u$ and $y = u + du$.

Draw thiis situation in two ways, showing (1) A top view of the xy plane, and (2) a three-dimensional view also showing the z -axis and the point P .

- (c) **[4 pts.]** What is the linear charge density of the strip?
- (d) **[9 pts.]** Recall from the previous week: an infinite thin rod of charge, if carrying linear charge density λ , produces a field

$$\frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

at a point which is at radial distance r from the rod. In the present geometry, what is the magnitude and direction of the element of field ($d\mathbf{E}$) produced at P by our infinitesimal strip of charge?

Hint: a picture of the yz plane, showing the strip position and the point P , is useful, and is expected.

- (e) [3 pts.] Find the z -component of $d\mathbf{E}$.

- (f) [**5 pts.**] We now want to use the principle of superposition to obtain the net field at P . This means summing over all infinitesimal strips to cover the complete plane of charge, i.e., integrating your expression from $u = -\infty$ to $u = \infty$. It might help to remember that

$$\int \frac{dv}{v^2 + 1} = \tan^{-1} v \quad \text{and} \quad \lim_{v \rightarrow \pm\infty} \tan^{-1} v = \pm \frac{\pi}{2}$$

Note that you only need to deal with the z -component; the other components are zero by symmetry.

- (g) [**2 pts.**] What is the dependence of the net field on z_0 ? Is this surprising?

2. Metallic sphere.

Metals are very good electric conductors. An ideal conductor conducts charges perfectly well. In a static situation, the electric potential is constant everywhere in the metal. All electric charges are forced to sit on the surface of the metal. (Otherwise the charges would move until this static situation is reached.)

We consider a solid metallic sphere of radius R , carrying a total charge Q .

- (a) [**3 pts.**] What is the surface charge density on the surface of the sphere?
- (b) [**6 pts.**] What is the electric field at distance $r > R$ from the center of the sphere? Use Gauss's theorem.
Your result should be the same as what you would get if all the charge was concentrated at the center of the sphere, instead of being spread out on the surface.
- (c) [**4 pts.**] What is the electric field at distance $r < R$ from the center, i.e., within the sphere? (We've mentioned already that the potential is constant at all points of the sphere.)
- (d) [**3 pts.**] What is the electric potential outside the sphere, $r > R$? Use the fact that the electric field is identical to a simpler situation.
- (e) [**4 pts.**] Sketch a plot of the electric potential V as a function of the distance r from the center of the sphere. Make sure to clearly mark the $r = R$ point.